Akash Kesav Shanmugaraj

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SUMMARY

Fourth-year Smart Manufacturing undergraduate at IIITDM Jabalpur focused on applied machine learning for materials science, with experience in distributed ML systems. Currently researching physics-informed inverse design, trust-minimized training, and semiconductor generative modelling; seeking laboratory internships to strengthen rigorous research experience and preparing for PhD study at leading universities worldwide.

EDUCATION

Indian Institute of Information Technology, Design and Manufacturing, Jabalpur — Expected May 2027
Bachelor of Technology in Smart Manufacturing | CGPA: **7.9 / 10** — Jabalpur, Madhya Pradesh
#1 IIIT; Govt.-funded Institute of National Importance

EXPERIENCE

Rakumura IT Solutions — Jun 2025 – Nov 2025
AI/ML Backend Development Intern (Remote) — Chennai, Tamil Nadu
- Architected **multi-agent AI systems** using **2 orchestration frameworks** (LangChain, LangGraph) and AI-assisted development tools (Claude Code, Codex, GitHub Copilot) for rapid prototyping, automating triage, assignment, and status tracking while **reducing manual coordination overhead** for internal operations teams
- Integrated a **4-system stack** (PostgreSQL, MongoDB, Redis, NATS) for asynchronous stateful execution across internal workflow endpoints; containerized services with Docker for reproducible deployments

IIITDM Jabalpur – Materials Science Lab — Nov 2023 – Apr 2024
Research Intern – Machine Learning (On-site) — Jabalpur, Madhya Pradesh
- Benchmarked **5 ML architectures** (GNNs, XGBoost, Random Forest, DNNs) on semiconductor band-gap prediction, achieving **83% accuracy**
- Designed an automated data-collection and preprocessing pipeline ingesting **1,600+ material records** from the Materials Project API across all 118 elements
- Contributed to research abstract accepted at **Materials Today Conference** (Elsevier, 2025)

CURRENT RESEARCH

Physics-Informed Inverse Design of Mechanical Metamaterials — Aug. 2026 – Present
Co-first-author research with Tejas P. Biradar; supervised by Dr. K. Ponappa — IIITDM Jabalpur
- Investigating inverse design of mechanical metamaterials under structural-integrity constraints using a Phys-JEPA-inspired latent formulation with PINN baselines; methodology and preliminary results remain confidential pending supervisor approval.

Trust-Minimized Hybrid Distributed Training over Untrusted Networks: Simulated Verifiable Coordination and Adaptive Trust Management — Feb. 2026 – Present
***Computing**, Springer (IF 3.6) — Under Review; equal-contributing author* — IIITDM Jabalpur
- Authors: Saksena*; **Akash Kesav***; Malpani; Parihar (*equal contribution). Co-designed a hybrid data/model-parallel framework with one-shot Krum filtering, softmax-weighted aggregation, VRF-and-trust validator selection, commit–reveal, and progressive cryptographic eviction.
- Built and evaluated ResNet-50/CIFAR-10 training experiments across data-parallel and pipeline configurations, including Byzantine attack scenarios, to validate the framework's robustness and overhead.

Semiconductor-Focused MatterGen Fine-Tuning and Evaluation — 2026 – Present
Confidential continuation of semiconductor inverse-design research — IIITDM Jabalpur
- Contributing to MatterGen fine-tuning for semiconductor candidates and building a systematic evaluation harness for reproducible checkpoint comparison; technical details withheld while experiments are finalized.

PUBLICATIONS & PATENTS

SECUF-2026 (Invited Talk) – Co-author, "Inverse design of magneto-excitonic semiconductors," 5th Conf. Physics under Synergetic Extreme Conditions (China).
Materials Today 2025 (Poster) – ML-based material property prediction, Sitges, Spain.
Patent Published – Co-inventor, automated scanning-based welding and additive-deposition system for path-edge restoration; Indian patent application No. 202621101529 (published 2026).
Patent Co-Filed – Co-inventor, portable non-destructive gold-purity estimator combining density and electrochemical measurements.

TECHNICAL SKILLS

Languages: Python, Go, SQL
Frameworks: PyTorch, PyTorch Geometric, TensorFlow, Scikit-learn, XGBoost, Hugging Face, LangChain, LangGraph
Core ML & AI: Deep Learning, GNNs, NLP, Transformers, RAG, Agentic AI
MLOps & Dev Tools: Docker, REST APIs, Go Fiber, NATS, Git, Linux, CI/CD, Claude Code, Codex, GitHub Copilot
Data & Databases: Pandas, NumPy, Matplotlib, Seaborn, PostgreSQL, MongoDB, Redis

LEADERSHIP

Team Lead, IIITDM Jabalpur – Led teams of 4–6 peers across **3 projects** in Product Realization, Engineering Design, and Cyber-Physical Production Systems; delivered prototypes through design reviews (grades **A, B+, A+**).